MATHEMATICS 2

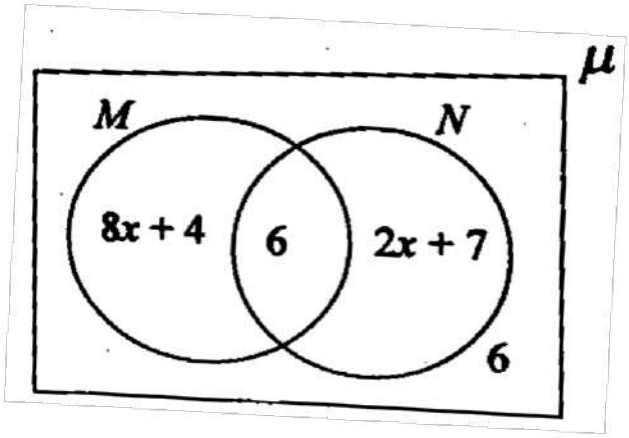
ESSAY

*Answer **four** questions **only**.*

All questions carry equal marks

*All working must be clearly shown. Marks will **not** be awarded for correct answers without corresponding working.*

1.



In the Venn diagram, M and N are intersecting sets in the universal set μ.

(a) Express n(M) and n(N) in terms of x.
(b) Given that n(M) = n(N), find the:
  i. value of x;
  ii. n(μ).
(c) Simplify: $2^6 \div (2^2 \times 2^1) \div 2^5$.

2. (a) Factorize the expression $5ay - by + 15a - 3b$.

(b) Solve $\frac{6}{4p-1} = \frac{4}{3(p+4)}$.

(c) Esi and Kofi shared an amount of GHC 21,00.00 in the ration 2:5 respectively. How much more did Kofi receive than Esi?

3. (a) If $r = \begin{pmatrix} -4 \\ -5 \end{pmatrix}$ and $m = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$, find **p** given that **p = r – m**.

(b) The sum of two numbers is 81. If the second number is twice the first, find the second number.

(c) The floor of a rectangular hall is of length of 9 m and width of 4 m. How many tiles of 20 cm by 30 cm can be used to cover the floor completely.

4. (a) Antwiwaa bought25 mangoes, 7 of which were unripe. What percentage of the mangoes were ripe?

(b)

x	1	2	3	4	... 8 ...	n
↓	↓	↓	↓	↓	↓	↓
y	−1	2	5	8	m	29

The mapping shows the relationship between x and y. Find the:

(i) rule of the mapping;

(ii) values of m and n.

(c) A bus left town **X** at 6:30 am and arrived at town **Y** at 1:00 pm. If the bus travelled at an average speed of 100km per hour, calculate the distance from town **X** to town **Y.**

5. (a) Simplify: $4x + 2)(x - 2) - 3x^2$.

(b) The following are the angles formed at the centre of a circle: $40°, 60°, 100°, 3x°$ *and* $5x°$. Find the value of x.

(c) The cost (C) in Ghana Cedis of producing a book of x pages is given by $C = 25 + 0.6x$.

(i) Find the cost of producing a book with 220 pages.

(ii) How many pages are in a book produced at a cost of GHC 145.00?

6. The table shows the number of marbles students sent to class for Mathematics lesson.

Number of Marbles (x)	*Number of Students (f)*	*fx*
1	4	-
2	5	-
3	-	42
4	9	-
5	-	3-
6	2	12

(a) Copy and complete the table.

(b) How many:

(i) students were in the class?

(ii) marbles were brought altogether?

(iii) marbles did **most** of the students bring?

(c) Calculate, correct to the **nearest** whole number, the mean number of marbles brought for the lesson.

END OF ESSAY TEST